

SUPPORT TICKET PRIORITY PREDICTION & AUTOMATED ASSIGNMENT

Intelligent Customer Incident Management System using Salesforce Agentforce

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1. EXECUTIVE SUMMARY & PROBLEM CONTEXT

Customer support ecosystems process hundreds of critical incident tickets daily. Traditional manual prioritization causes critical bottlenecks, leads to delayed triage for mission-critical failures, and lowers overall Customer Satisfaction (CSAT). This project implements an autonomous conversational intelligence framework that:

Autonomous Classification

Evaluates incoming case descriptions to dynamically tag tickets as High, Medium, or Low priority.

Smart Routing & Tasking

Automatically routes critical issues to Senior Support Specialists and spawns escalation tasks.

Agentforce Integration

Natural language conversational interface capable of retrieving context and triggering flows.

SLA Protection

Monitors turnaround thresholds and triggers automated flags to prevent deadline breaches.

2. TARGET CUSTOM OBJECT SCHEMA

Data persistence and triage logs are structured inside the `Support_Ticket_Intelligence__c` custom Salesforce object:

Field Label	API Name	Data Type	Description & Attributes
Ticket Number	Ticket_Number__c	Auto Number	Formatted auto-identifier: TKT-{0000}
Customer	Customer__c	Lookup (Account)	Parent account associated with the ticket
Contact	Contact__c	Lookup (Contact)	Designated customer point of contact
Issue Type	Issue_Type__c	Picklist	Values: Technical, Billing, General
Description	Description__c	Long Text Area	Detailed incident and operational logs
Priority Level	Priority_Level__c	Picklist	Triage tier: Low, Medium, High
Status	Status__c	Picklist	New, In Progress, Resolved
Assigned To	Assigned_To__c	Lookup (User)	Assigned tier or engineer
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Risk trigger flag for breach prevention

3. AUTO-LAUNCHED FLOW ORCHESTRATION LOGIC

The backend logic runs on an Auto-Launched Flow (Support_Ticket_Intelligence) invoked by Agentforce upon user intent recognition. It processes input parameters, classifies ticket urgency via text analysis, and provisions escalation tasks.

Flow Interface Contract:

- **Input:** varAccountName (Text) – Captures account entity name from conversational prompt.
- **Outputs:** varAccountId, varTicketId, varPriorityLevel, varAssignedTo, varActionMessage.

Step 1 & 2: Entity Lookup & Variable Assignment

Get Records & Assignment

Queries Account where Name Equals {!varAccountName} (Sorted: CreatedDate DESC, First record only). Stores {!Get_Account.Id} into varAccountId.

Step 3 & 4: Retrieve Latest Incident Record

Get Records & Assignment

Fetches Support_Ticket_Intelligence__c where Customer__c Equals {!varAccountId} (Sorted: CreatedDate DESC). Assigns ticket ID to varTicketId.

Step 5 & 6: Keyword Heuristics & Priority Decision

Decision Engine

- **High Priority:** Contains "urgent", "not working", or "failure" → Sets varPriorityLevel = "High"
- **Medium Priority:** Contains "issue", "slow", or "delay" → Sets varPriorityLevel = "Medium"
- **Default Outcome:** Sets varPriorityLevel = "Low"

Step 7, 8 & 9: Urgent Case Tasking & Agent Assignment

Create Records & Assignment

If varPriorityLevel == 'High': Generates urgent escalation Task record (Subject: "Urgent Ticket Handling", Priority: High, Status: Not Started). Populates varAssignedTo = "Senior Support Agent".

Step 10 & 11: SLA Verification & Conversational Payload

Assignment

Calculates ticket age (> 2 days flags SLA breach). Prepares localized feedback message for Agentforce response display based on classified priority.

4. FLOW ARCHITECTURAL PATH

```
[Start: Agentforce Action] → Get Account → Assign (varAccountId) → Get Support Ticket → Assign (varTicketId)
  → Decision: Analyze Description Keywords
    └─ [High Priority] → Create Escalation Task → Assign Senior Agent → SLA Check → Assemble Response
    └─ [Medium Priority] → Queue for Assignment → SLA Check → Assemble Response
    └─ [Low Priority] → Standard Queue → SLA Check → Assemble Response → [End / Return to Agentforce]
```

5. AGENTFORCE SUBAGENT & TOPIC CONFIGURATION

Salesforce Agentforce is configured as the conversational orchestrator, interpreting natural language input and executing the priority prediction pipeline.

Configuration Item	Implementation Details
Topic Name	Support Ticket Priority Analysis (API: Support_Ticket_Priority_Analysis)
Classification Description	Analyzes customer support tickets to determine priority based on issue description and urgency. Handles inquiries regarding ticket status, issue severity, and support assignment.
Agent Scope	Strictly restricted to analyzing ticket descriptions, determining priority (High, Medium, Low), and triggering flow automation for task creation. Excludes billing and user account management.
Directives / Instructions	1. Prompt for Account Name if absent from conversation. 2. Fetch the latest ticket linked to the account and evaluate narrative urgency. 3. Trigger Flow Support_Ticket_Intelligence and return formatted ticket resolution summary.

6. AGENT ACTION PARAMETER MAPPING

Parameter	Type	Direction	Agent Interaction Behavior
varAccountName	Text	Input	Prompted / Collected from user during conversation
varAccountId	Text	Output	Returned in conversation (Account 18-digit identifier)
varTicketId	Text	Output	Displays unique ticket record ID
varPriorityLevel	Text	Output	Displays calculated priority tier (High, Medium, Low)
varAssignedTo	Text	Output	Displays designated engineering tier / agent
varActionMessage	Text	Output	Final automated summary narrative delivered to customer

7. VERIFICATION & PRACTICAL IMPACT

Validation Walkthrough: When the operator queries: *"give me my supportticket data"*, Agentforce matches the user intent with the Support Ticket Priority Analysis topic, triggers the background Flow, scans ticket TKT-0002 containing the description keyword *"urgent"*, tags the case as **High Priority**, assigns a Senior Support Agent, creates an immediate follow-up Task, and returns complete structured metadata to the console within seconds.